

Josiah Myhre

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Portfolio: josiahmyhre.com

Mechanical engineering graduate with Engineer in Training (EIT) certification application in progress, and over 12 years of experience in product configuration, manufacturing support, fixture design, documentation, and process improvement seeking permanent full-time employment as an engineer.

Education

The University of Texas at Dallas – B.S. in Mechanical Engineering, Cum Laude (3.79 GPA)	August 2026
Winona State University – B.A. in Social Work	2010

Certifications

EIT certification application submitted	August 2026
Fundamentals of Engineering (FE) Mechanical Exam – Passed (Jan 2026)	

Selected Technical Coursework

Introduction to Programming (C) • Programming for Engineers (C++) • Circuit Analysis I • Circuits and Applied Electronics
• Introduction to Computational Design & Analysis (FEA using Abaqus) • Additive Manufacturing

Engineering Design & Project Experience

Senior Design – Modular Rail Drilling Machine [\[Project Page\]](#)

Technical details withheld due to the sponsor's intent to file a patent

- **Outstanding Project Award, UT Dallas Senior Design Expo**
- Designed a mechanical system to meet sponsor requirements through detailed design and prototyping
- Performed stress, deflection, FEA, gear-train, and circuit analyses to validate design decisions
- Fabricated and assembled a functional prototype using in-house machined and 3D-printed components
- Developed and executed a quantitative test plan to validate prototype performance
- Iteratively tested and redesigned components using load and moment analysis to resolve mechanical issues
- Integrated mechanical, electrical, and instrumentation subsystems into a functional prototype
- Produced mechanical drawings, electrical schematics, 3D models, and control software documentation

Independent Project – Autonomous Profiling Float (In Progress) [\[Project Page\]](#)

- Designed and prototyped an autonomous profiling float using variable displacement for depth control
- Integrated an ESP32, pressure sensor, encoder, limit switch, and servo-driven lead-screw actuator
- Implemented embedded code for homing, position tracking, actuator control, and limit detection
- Implemented encoder-based position tracking to monitor lead-screw displacement
- Developed pressure-feedback control logic for autonomous depth adjustment
- Designed a PCB, integrating sensor interfaces, actuator connections, and power distribution
- Bench-tested mechanical, electrical, and control subsystems before full system integration
- Iterated multiple prototypes to improve packaging, cable management, accessibility, and assembly
- Performing buoyancy and displacement calculations for neutral buoyancy and depth control targets

Robosub | UTD Water Robotics Club

Subformica — 2025 MATE ROV Competition | Mechanical Team Lead [\[Project Page\]](#)

- Led mechanical design of a competition ROV from concept through fabrication to deployment
- Coordinated mechanical design tasks, subsystem interfaces, and fabrication activities across the project team
- Designed and machined a modular chassis with vectored thrusters for six-degree-of-freedom motion

- Designed a two-axis differential mechanism providing independent 90° manipulator tilt and rotation
- Selected differential gearing to translate 180° servo travel into the required manipulator articulation
- Designed a servo-actuated rack-and-pinion system to extract 50 mL samples from sealed containers
- Developed tool-less, quick-change manipulator attachments to expand task capability
- Calculated required buoyancy and fabricated foam panels to achieve near-neutral buoyancy
- Prototyped and tested an adjustable camera mount to establish final positioning and geometry
- Machined and pressure-tested a watertight 6061 aluminum/acrylic electronics enclosure
- Fabricated vehicle hardware in-house using manual machining and additive manufacturing
- Coordinated with electrical and programming teams on packaging, actuator limits, and tooling logic

VESPA — 2024 MATE ROV Competition | Mechanical Team Member [\[Project Page\]](#)

- Contributed to a competition ROV that placed 3rd overall and 1st in technical documentation
- Designed a pneumatic manipulator for grasping, centering, and retaining competition task objects
- Developed a servo-actuated retrieval system with a helically grooved spool for controlled winding
- Designed a self-centering servo-driven valve rotation tool tolerant of vehicle misalignment
- Fabricated and tested mechanical tooling through multiple iterations before competition
- Designed a serviceable topside enclosure integrating power, pneumatics, networking, safety, and cooling

LUNA — 2025 RoboBoat ASV Competition | Mechanical Team Support [\[Project Page\]](#)

- Designed and fabricated a gyroscopic gimbal to stabilize stereo-camera and LiDAR sensors for an autonomous boat
- Positioned one gimbal axis through the center of mass and the second in its plane to minimize disturbance torque
- Repurposed an HDD BLDC motor and stacked platters as a compact high inertia flywheel
- Integrated the gimbal with existing sensor hardware, wiring, and vehicle interfaces
- Tested the prototype and traced residual oscillation to insufficient flywheel angular momentum
- Compared flywheel-on and flywheel-off behavior under induced disturbances to assess stabilization effectiveness
- Delivered the gimbal on a compressed schedule to support a short-staffed mechanical team

Work Experience

Ergomart – Dallas, TX

Engineering Support Specialist

Jan 2025 – Present

- Designed jigs and fixtures to improve assembly repeatability, accuracy, and production consistency
- Resolved shop-floor issues with engineering and production to improve manufacturing readiness and quality
- Created assembly instructions, installation guides, and technical documentation for production and field use

Customer Service and Sales Manager

Sept 2017 – Jan 2025

- Managed a 4–6 person customer service and sales team, setting priorities and standardizing workflows
- Led cross-functional coordination between sales, engineering, production, and shipping to ensure order accuracy
- Served as the technical interface between customers, engineering, and production, to meet customer requirements
- Coordinated purchasing and vendor issues, maintaining production schedules to meet customer deadlines

Assembly Team Lead

Sept 2013 – Sept 2017

- Implemented an inventory and part-tracking system by organizing components and coordinating ERP workflows
- Represented assembly in design reviews, providing DFM/DFA feedback to improve manufacturability/assembly
- Led assembly support, training new team members on procedures, quality expectations, and product builds
- Supported hospital equipment installations, troubleshooting assembly and fitment issues in the field

Skills

Design Tools: Creo, SolidWorks, Fusion, Blender

Manufacturing/Design: Jigs and fixtures, CAD drawings, BOM support, DFM, DFA, manual machining, FDM, resin casting

Electrical/Electronics: PCB design, soldering, electrical schematics, circuit analysis, microcontrollers, I/O integration

Tools/Analysis: Excel, NetSuite ERP, Jira, Notion, Google Workspace, MATLAB, Simulink, Mathcad